

Redesign Spec & Implementation Playbook

Jobs scope discipline × Musk sequencing discipline, applied to the LSB build. GO/NO-GO at week 3. Live-ready at week 13. Every validation gate bit-identical to Master Blueprint v2.1.

Delete everything that doesn't answer the edge question. Interrogate every requirement and parallelize everything the calendar allows. Treat the forward test and trade-count gates as launch windows that no ambition compresses.

Supersedes: sequencing, module plan, and session structure of Master Blueprint v2.1 · **Inherits unchanged:** eight entry gates, all strategy logic, risk parameters, pessimistic fill model (Part 7.1), seven-table DDL (Part 5), GO/NO-GO thresholds (Part 14), ADR-001, instrument list (Appendix A), risk register · **Date:** June 2026 · **Status:** Active — Phase A unlocked, Phases B/C locked

The Two Mechanics, Translated

The product is not a trading platform. The product is a validated answer to one question: does the liquidity sweep edge survive walk-forward validation?

1.1 Jobs (the 90-day product cut) — a scope weapon

Jobs didn't speed Apple up; he deleted roughly 70% of the product line by asking what the company existed to do. The LSB translation: every module, table, state, and tool that does not serve the edge question is deferred or deleted until after the answer is GO. The Jobs 2x2 for the LSB has two columns — **prove the edge** and **trade the edge** — and Phase A builds only the first.

1.2 Musk (the 19-day migration) — a sequencing weapon

The migration worked because every requirement was interrogated until only physics remained, and everything parallelizable ran in parallel. The algorithm, in strict order: **(1)** make the requirements less dumb, **(2)** delete the part or process — if you don't add ~10% back later, you didn't delete enough, **(3)** simplify and optimize, only after deleting, **(4)** accelerate cycle time — never accelerate what should have been deleted, **(5)** automate, last.

The non-negotiable corollary — the Physics Rule. Musk compresses engineering, not physics; SpaceX does not negotiate with launch windows. The LSB's physics: market data arrives at 1x speed, and statistical conclusions require trade count. The 8-week demo forward test and trade-count-gated outcome learning are launch windows — protected and non-compressible, exactly as declared in v2.1. This document moves aggression *around* them, never through them. The v2.1 risk register names schedule pressure eroding validation gates as the top open risk; this redesign is engineered so all schedule aggression lands on build scope and sequencing while both gates remain bit-identical.

1.3 Requirement interrogation (Step 1: "less dumb")

v2.1 requirement	Who needs it, and when?	Verdict
14 modules (M1–M14)	Only ~5 needed to answer the edge question	Cut to 5 for Phase A ; rest deferred
7-table PostgreSQL schema	Live-trading tables serve no purpose pre-GO	Cut to 4 tables for Phase A
9-state operating machine	A backtest has no broker, no orders, no halts	0 states Phase A; 5 Phase B ; remaining 4 must re-justify existence
8 entry gates	This is the strategy under test	Untouchable — substance, not scope
29-session playbook	Sized for the full platform, serialized	Re-cut: 12 sessions to GO ; remainder parallelized
Walk-forward validation (18mo/6mo, ≥3 instruments)	The validation gate itself	Untouchable
8-week forward test	Physics (calendar time)	Untouchable — but no longer dead time (\$1.6)
Trade-count-gated learning	Physics (sample size)	Untouchable
Graphify knowledge layer	Valuable — but does it move the GO date?	Defer seeding to Phase B (Musk 10% rule candidate)
Docker / VPS	Needed only for unattended live operation	Phase B parallel track , deployed Phase C
MT5 integration	Needed for forward test, not backtest	Phase B, first item

1.4 The cut (Step 2: delete)

Modules: M1–M14 → 5 core modules (Phase A)

Survives Phase A	Function
C1 — Data Pipeline	Historical candle ingestion, integrity checks, gap handling (absorbs the v2.1 data modules)
C2 — Signal Engine	The 8 entry gates, exactly per v2.1 spec — pure functions, no broker awareness
C3 — Backtest Engine	Deterministic, event-driven, pessimistic fill simulation per v2.1 Part 7.1 (spread, ATR-scaled slippage, gap-through-stop fills, commission/swap)
C4 — Walk-Forward Harness	Rolling 18-month train / 6-month test windows, ≥3 instruments, automated window orchestration
C5 — Verdict Report	Metrics, equity curves, per-window stability, machine-rendered GO/NO-GO verdict against v2.1 Part 14 thresholds

Deleted from Phase A (deferred, not destroyed): broker abstraction, order management, live state machine, monitoring/alerting, dashboard/UI, deployment tooling, notification layers, Graphify seeding. Every item is either rebuilt in Phase B's parallel track or interrogated again before being added back.

Schema: 7 tables → 4 (Phase A)

`config_version` · `candle` · `signal` · `wf_run` (with child `sim_trade`). The three live-trading tables of v2.1 Part 5 are restored **verbatim** in Phase B — their DDL already exists; deferral costs nothing.

State machine: 9 → 0 → 5

Phase A: a backtest is a pure function; it has no operating states. Phase B needs five: `IDLE` → `SCANNING` → `PENDING_ENTRY` → `IN_TRADE` → `HALTED` . Before Phase C, each of the four remaining v2.1 states must justify its existence with a concrete failure scenario it uniquely handles, recorded as an ADR — or it stays deleted.

1.5 Phase A — "Prove" (weeks 1–3)

Single deliverable: the Verdict Report. GO/NO-GO moves from week 7 to end of week 3. The compression is real, not cosmetic: v2.1's week-7 gate carried platform scaffolding alongside the backtest build; with everything non-essential deleted, the remaining work is twelve Claude Code sessions (Part II, §2.2), with the data lane and the gates lane running in parallel. **Gate criteria are unchanged from v2.1** — same thresholds, same multi-instrument requirement, same per-window stability tests. The gate moved; the bar did not. **A NO-GO is a success outcome**: the answer arrived in 3 weeks instead of 7 at near-zero infrastructure cost, which is the entire point of this sequencing.

1.6 Phase B — forward test + parallel hardening (weeks 4–13)

This is the Musk move. v2.1 implicitly serializes validate → forward test → build live infrastructure, but the forward test is **calendar time, not engineering time** — the system trades demo while the build continues beside it. Track 1 runs the frozen 8-week forward test (launch week 5, after MT5 bridge, 5-state machine, live-loop wiring, and a 48-hour dry run). Track 2 builds the hardening layer in parallel: remaining v2.1 tables, risk/portfolio modules, monitoring, Docker, VPS shadow deployment, state-machine interrogation, and Graphify seeding. This preserves the v2.1 principle — validation before infrastructure spend — because Track 2 begins only after the week-3 gate passes, and nothing touches real capital pathways until the forward-test gate. What it eliminates is eight weeks of idle engineering time.

1.7 Phase C — "Scale" (week 13+)

Promote the burned-in Track 2 build to the VPS, then apply Musk's 10% rule deliberately: review everything deleted in §1.4 and add back only what was genuinely missed, with a written ADR per item. Hiring triggers remain tied to revenue milestones per v2.1.

1.8 Timeline comparison

Milestone	v2.1	v3.0	Mechanism
GO/NO-GO (backtest gate)	Week 7	Week 3	Jobs: scope deletion
Forward test complete	~Week 15–20 (serialized)	Week 13	Musk: parallelization
Forward test duration	8 weeks	8 weeks	Physics — untouched
Outcome-learning gates	Trade-count	Trade-count	Physics — untouched
Live-ready	~Week 20+	Week 13	Both

Roughly seven to eight weeks reclaimed. **Zero weeks taken from any validation gate.**

1.9 The honesty clauses (carried forward, reinforced)

- Schedule pressure eroding validation gates is the named top risk; both gates' criteria live in spec config read by the verdict code, so softening the bar requires a visible diff to a protected file, reviewable as an ADR.
- Speed of construction says nothing about existence of edge. A NO-GO at week 3 or week 13 is the system working as designed.
- Most retail algorithmic strategies lack durable edge; nothing in this document is investment advice or a performance projection.
- "Quantum-inspired" continues to mean classical Bayesian belief-state mechanics, stated plainly wherever it appears.

Sessions, Gates, and Calendar

Twelve sessions to GO/NO-GO, two parallel tracks through the forward test, three promotion sessions. Every session: a copy-pasteable Claude Code prompt and explicit accept-when criteria.

2.1 Working agreements (all phases)

Determinism in CI from session A6 onward: the golden-fixture replay runs twice per commit; both runs must produce identical output hashes.

Config integrity: YAML per instrument → frozen dataclass → canonicalised → sha256 `config_hash` → insert-or-get into `config_version`. Identical configs always produce identical hashes; every result row references the `config_hash` that produced it. Walk-forward results are meaningless without this.

Spec immutability: the v2.1 strategy spec is immutable during Phase A. Proposed strategy changes are written to `docs/decisions/` and parked until after the verdict. Changing the strategy while validating it invalidates the validation.

Session hygiene: each session ends with green CI and a commit. No accept-when criteria met, no merge.

2.2 Phase A sessions — "Prove" (weeks 1–3)

Phase A schema — migration `001_core.sql`

`config_version` (`config_hash` PK, canonical YAML, `created_at`) · `candle` (`instrument`, `timeframe`, `ts`, OHLCV, `spread`; unique on `instrument+timeframe+ts`) · `signal` (per-candle evaluation: each of the 8 gates pass/fail with reasons, FK to `config_version`) · `wf_run` (one row per walk-forward window: `instrument`, `train span`, `test span`, `config_hash`, metrics JSONB) with child `sim_trade` (`entry/exit`, `prices`, `slippage applied`, `R-multiple`, FK to `wf_run`).

SESSION A0 • Scaffold — week 1, day 1

```
Create the reduced repository structure: src/lsb/
{data,signals,backtest,walkforward,report}/, config/, migrations/, tests/, docs/
decisions/, scripts/. Write docs/PHASE_STATUS.md marking Phase A active and Phases B/C
locked. Carry forward docs/decisions/ADR-001-broker-bridge.md (native-Windows MT5, EA-
socket fallback, status: accepted — implementation deferred to B1). Set up pytest and CI
running unit tests on every commit. Do NOT create broker, state-machine, Docker, or
deployment directories — they do not exist in Phase A.
```

Accept when: tree matches; CI green (zero tests is fine today); PHASE_STATUS explicitly locks B/C.

SESSION A1 • Config system + schema — week 1

Implement the config system: YAML per instrument in config/, frozen dataclass, canonicalisation, sha256 config_hash, insert-or-get into config_version. Write migrations/001_core.sql with the four Phase-A tables, apply to the dev database, write round-trip tests.

Accept when: identical configs → identical hashes, always; four tables exist; round-trip tests green.

SESSION A2 • Data pipeline, instrument #1 — week 1 • || lane 1

Write scripts/fetch_history.py pulling ≥3 years of H1 OHLCV for the first Appendix-A instrument (broker export or Dukascopy) into Parquet under data/history/, then scripts/audit_history.py reporting gaps, duplicate timestamps, weekend bars, DST anomalies, and spread outliers. Fail the audit on any unexplained gap >2 candles. Produce docs/data_audit_report.md. Load audited data into candle.

Accept when: audit report exists and every flagged anomaly has a written disposition.

SESSION A3 • Data pipeline, instruments #2-#n — week 1 • || lane 1

Extend fetch/audit/load to the full Appendix-A list (≥3 instruments per the walk-forward requirement). One audit report per instrument, same dispositions rule.

Accept when: ≥3 instruments fully audited and loaded; per-instrument candle counts reconcile with expected trading hours.

SESSIONS A4-A5 • Signal engine: the eight gates — weeks 1-2 • || lane 2 (parallel to A2-A3; no shared code)

A4: Implement entry gates 1-4 exactly per Blueprint v2.1, as pure functions over candle windows: no broker awareness, no side effects, no clock. Build each gate test-driven against hand-labeled fixtures – per gate, at least one fixture that should pass and two near-misses that should fail, labeled by hand from real chart data before the gate is coded. Every evaluation writes a signal row with per-gate pass/fail and reasons.

A5: Gates 5-8, same discipline. Then the conjunction: a setup qualifies only when all eight gates pass on the same evaluation, per v2.1. Add property tests: shuffled candle history must never qualify at a rate comparable to real history.

Accept when: all 8 gates green against hand-labeled fixtures; near-misses correctly rejected; conjunction matches the v2.1 spec verbatim; the shuffle property test passes.

SESSIONS A6–A7 • Backtest engine — week 2

A6: Implement the event-driven backtest core: injectable Clock (replay mode), per-candle evaluation loop streaming Parquet history through the signal engine, position lifecycle (pending → filled → managed → closed) per v2.1 trade-management rules.

A7: Implement SimulatedBroker with the v2.1 Part 7.1 pessimistic fill model in full: spread costs, ATR-scaled slippage, gap-through-stop fills at the gapped price, commission and swap. Unit-test from worked examples — each cost component verified against a hand-computed expected fill.

Accept when: a sample month replays end-to-end producing signal rows for 100% of candles; every fill in the test set matches its hand-computed value; no fill is ever better than the pessimistic model allows.

SESSION A8 • Determinism gate — week 2

Wire the golden-fixture replay into CI: replay a fixed sample month twice per commit; both runs must produce identical output hashes (signals, trades, equity). Any nondeterminism (dict ordering, float accumulation order, timestamps in output) is a build failure. Document the determinism contract in docs/determinism.md.

Accept when (Gate GA-1): two consecutive CI runs, identical hashes; an intentionally injected nondeterminism is caught by the gate.

SESSION A9 • Walk-forward harness — week 3

Implement src/lsb/walkforward/: rolling 18-month train / 6-month test windows per v2.1, orchestrated across all loaded instruments. Each window runs the deterministic engine, persists one wf_run row with full metrics and child sim_trade rows, keyed to config_hash. Windows are independent — run them in parallel processes.

Accept when: full window set completes for ≥3 instruments; every wf_run is reproducible (re-run → identical metrics); no window's test span overlaps its own train span.

SESSION A10 • Verdict report — week 3

Implement src/lsb/report/: per-window and aggregate metrics, equity curves, drawdown profiles, per-window stability analysis (sign and dispersion across windows and instruments), and a machine-rendered GO/NO-GO verdict evaluated strictly against the Blueprint v2.1 Part 14 thresholds — thresholds read from the spec config, never re-typed. Output: docs/verdict_report.md plus rendered charts.

Accept when: report renders from a real run; verdict logic unit-tested against synthetic pass/fail metric sets; no threshold hard-coded outside the spec config.

SESSION A11 • Full run + verdict — week 3, final

Execute the complete walk-forward across all instruments on audited data. Review every anomaly in docs/anomaly_review.md: outlier windows, suspicious fills, gate-pass-rate drift across time. Render the Verdict Report. Record the verdict in docs/decisions/ADR-002-go-no-go.md.

GATE GA — GO/NO-GO (end of week 3). Criteria: Blueprint v2.1 Part 14, unchanged — multi-instrument requirement, per-window stability tests, aggregate thresholds. The gate moved from week 7 to week 3; the bar did not move. **On NO-GO:** stop; no Phase B work begins. Archive the run, write the post-mortem, and treat the parked docs/decisions/ ideas as hypotheses for a new ~3-week Phase A cycle — or retire the strategy. A NO-GO at week 3 having cost zero infrastructure is this plan working exactly as designed.

2.3 Phase B — forward test + parallel hardening (weeks 4–13, conditional on GO)

Track 1 — the launch window (sessions B1–B4, then 8 weeks of market time)

SESSION B1 • MT5 demo bridge — week 4

Implement the broker bridge per ADR-001: native-Windows MT5, EA-socket fallback. It must conform to the same broker interface SimulatedBroker implements — write a conformance test suite that both brokers pass identically. Demo account only; the live-account code path does not exist yet.

Accept when: conformance suite green on both brokers; order round-trip verified on demo.

SESSION B2 • Five-state operating machine — week 4

Implement the reduced operating machine: IDLE → SCANNING → PENDING_ENTRY → IN_TRADE → HALTED. Every transition writes a state_transition row with reason. HALTED is reachable from every state on: data integrity failure, broker disconnect, daily-loss limit per v2.1 risk parameters, or manual command — and requires manual acknowledgment to exit. Unit-test all legal and illegal transitions.

Accept when: full transition matrix tested; HALTED behavior verified for all four trigger classes.

SESSION B3 • Live loop wiring — week 5

Wire the (unchanged) signal engine and trade management into the live evaluation loop with the Clock in live mode and the MT5 demo bridge. Snapshot logging for 100% of evaluated candles, same signal schema as backtest — forward-test data must be directly comparable to walk-forward data. Add a daily integrity audit job (gaps, spread anomalies, clock drift) that transitions to HALTED on failure.

Accept when: 48-hour dry run on demo completes with 100% candle coverage, zero unexplained gaps, zero unhandled exceptions.

SESSION B4 • Forward test launch — end of week 5

Launch checklist: config_hash frozen and recorded; Gate GB thresholds written down **before** the test starts (pre-registration); daily audit green for the prior 48h. Then start the clock.

The freeze rule (weeks 5–13). No parameter changes, no gate-logic changes, no "small fixes" to strategy code for the duration. Infrastructure defect fixes (logging, reconnects) are allowed with a written entry in docs/forward_test_log.md ; anything touching signal or trade-management logic restarts the 8-week window. Weekly review is observation only. No early termination on good results; no extension-as-rescue on bad ones.

Track 2 — hardening (sessions B5–B10, weeks 5–11, parallel; deployed only after Gate GB)

Session	Build
B5	Restore the remaining v2.1 Part 5 tables verbatim (migrations/002_live.sql); round-trip tests.
B6	Risk and portfolio modules per the v2.1 module spec (the deferred M-modules), test-driven against v2.1 risk parameters — position sizing, exposure caps, daily-loss enforcement feeding the HALTED trigger.
B7	Monitoring and alerting: heartbeats, state-transition alerts, daily digest. Alert on silence, not just on errors.
B8	Docker images for the full system; compose file; image build in CI.
B9	VPS provisioned and security-hardened — <i>provisioned, not promoted</i> . The forward test stays where it is; the VPS runs a shadow deployment against demo to burn in.
B10	Two interrogations. (a) The four deleted v2.1 states: each returns only with a written concrete failure scenario it uniquely handles, recorded as an ADR — else it stays deleted. (b) Graphify seeding: <code>pip install graphify</code> (double y), <code>graphify install --platform claude , /graphify .</code> at repo root with <code>/docs</code> holding the v2.1 blueprint, this v3.0 document, and the spec set; <code>--postgres</code> introspection now that the full schema exists; commit <code>graphify-out/</code> . Day-to-day: <code>graphify update .</code> after merges, <code>graphify affected</code> before any risk-math change, <code>--force</code> after file deletions.

GATE GB — forward-test verdict (week 13). Criteria per v2.1: the forward-test trade distribution compared against walk-forward expectations — distribution comparison, not cherry-picked headline metrics. Trade-count sufficiency applies: if the 8 weeks produced fewer trades than the v2.1 minimum, the verdict is **INSUFFICIENT — extend in whole-week increments until the count gate is met**. Trade count is physics; the calendar serves it, not the reverse. **On NO-GO:** the VPS is not promoted, no live account is opened, and the divergence analysis (fills? regime? gate pass rates?) becomes the input to a new Phase A cycle. Track 2 work is retained — it was built against the immutable broker interface and is strategy-agnostic.

2.4 Phase C — "Scale" (week 13+, conditional on GB GO)

SESSION C1 • Promotion

Cutover checklist: VPS shadow deployment has ≥ 2 weeks clean burn-in; alert paths verified end-to-end (kill the process, confirm the alert arrives); HALTED → manual-acknowledge flow drilled once deliberately. Promote to VPS, still on demo, for one final week. Then — only on explicit manual decision, never automated — minimum-size live per the v2.1 capital schedule.

SESSION C2 • The 10% review

Walk the full §1.4 delete list. Each candidate returns only with a written justification ADR. If nothing returns, the deletions were correct; if more than ~10% returns, the Phase A cuts were too shallow — record that for the next project.

SESSION C3 • Outcome learning activation

Trade-count-gated exactly per v2.1: the learning loop activates only when live trade count crosses the v2.1 minimum; updates batched per the spec, never per-trade. Hiring triggers remain tied to v2.1 revenue milestones.

2.5 Calendar

Week	Track	Milestone
1	A0-A3 A4-A5	Scaffold, config/schema, data audited (≥ 3 instruments) gates 1-8 fixture-tested
2	A6-A8	Backtest engine, pessimistic fills, Gate GA-1: determinism in CI
3	A9-A11	Walk-forward, Verdict Report, Gate GA: GO/NO-GO
4	B1-B2	MT5 demo bridge, 5-state machine
5	B3-B4	Live loop, 48h dry run, forward test launch + freeze
5-11	T1: market time T2: B5-B10	Forward test running live tables, risk modules, monitoring, Docker, VPS shadow, state interrogation, Graphify
12-13	T1 concludes	Eight full weeks of market time complete
13	—	Gate GB: forward-test verdict (extend by whole weeks if trade count insufficient)
13+	C1-C3	VPS promotion, 10% review, trade-count-gated learning

v2.1 reached this point at roughly week 20+; v3.0 reaches it at week 13. The reclaimed ~7-8 weeks came from scope deletion (weeks 1-3 vs 1-7) and parallelization (weeks 5-11 doing double duty). **Weeks taken from validation: zero.**

2.6 Risk register deltas

The v2.1 register carries forward in full. Three entries updated or added:

Risk	Mitigation
Schedule pressure eroding validation gates (existing — top risk)	Strengthened: both gates' criteria live in spec config read by the verdict code; softening the bar requires a visible diff to a protected file, reviewable as an ADR.
Parallel-track contamination (new)	The forward test runs from a tagged release on isolated infrastructure; Track 2 develops on main; nothing deploys to the forward-test host during the window.
Compression spillover (new)	The momentum of a 3-week GO tempting compression of market time. The freeze rule and trade-count extension rule are written into the gate definition itself, and the launch checklist requires GB thresholds recorded <i>before</i> the test starts — pre-registration, so the bar cannot drift after seeing results.

Standing disclaimers unchanged: most retail algorithmic strategies lack durable edge; a GO at either gate is evidence, not a guarantee; nothing herein is investment advice; "quantum-inspired" means classical Bayesian belief-state mechanics throughout.

2.7 Definition of done, per phase

Phase A done: Verdict Report rendered from a deterministic, multi-instrument walk-forward on audited data; verdict recorded as ADR-002. **Phase B done:** 8+ weeks of frozen forward test meeting the trade-count gate; GB verdict recorded; hardened stack burned in on VPS shadow. **Phase C done:** promoted system live at minimum size per the v2.1 capital schedule; 10% review complete; outcome learning armed behind its trade-count gate.

The plan gets to every answer faster, and to no answer early.
